

AlphaGenome Computational Analysis

TTN Pathogenic Deletion — Cardiac Tissue Expression Modeling

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IMPORTANT: This is not a clinical diagnostic report.

These visualizations were generated using Google DeepMind's AlphaGenome model, a research-grade computational tool for predicting the genomic effects of sequence variants. Results represent model predictions, not functional laboratory data. They are included here as supplementary context for specialist review alongside confirmed clinical findings.

Variant Information

Gene	TTN (Titin)
HGVS Notation	c.73318_73324delGAATGCT
Protein Effect	p.E24440MfsX2 (frameshift, premature stop)
Zygosity	Heterozygous
Classification	Pathogenic
Genome Build	hg38 — chr2:178567885
Tissue Modeled	UBERON:0000948 (Heart)
Analysis Window	1 Mb centered on variant (chr2:178,043,600 – 179,092,176)

Genomic Context: Why This Deletion Matters

TTN encodes titin, the largest protein in the human body and the primary elastic component of the cardiac sarcomere. Titin functions as a molecular spring, maintaining sarcomere integrity during contraction and relaxation cycles. Truncating variants in TTN are the single most common genetic cause of dilated cardiomyopathy (DCM), accounting for approximately 25% of familial cases.

This variant involves the deletion of 7 base pairs (GAATGCT). Because 7 is not divisible by 3, the deletion causes a frameshift — the entire downstream reading frame is scrambled, resulting in a premature stop codon at position E24440 (p.E24440MfsX2). The truncated titin protein cannot maintain sarcomere structure under mechanical load, which is the mechanistic basis for the observed dilated cardiomyopathy phenotype in this family.

Family History Context

This variant demonstrates high penetrance across generations. The patient's father developed progressive systolic heart failure and died from this disease. The patient's brother carries the same variant and is further along in disease progression. The patient's current EF trajectory (56% → ~50%) is consistent with the early-to-moderate phenotypic expression pattern seen in this family.

AlphaGenome Output: Combined Panel Overview

The figure below shows all modeled outputs across the 1 Mb TTN locus for heart tissue. The vertical orange line marks the variant position (chr2:178567885). REF (reference/normal sequence) and ALT (deletion/variant sequence) tracks are overlaid in gray and red respectively.

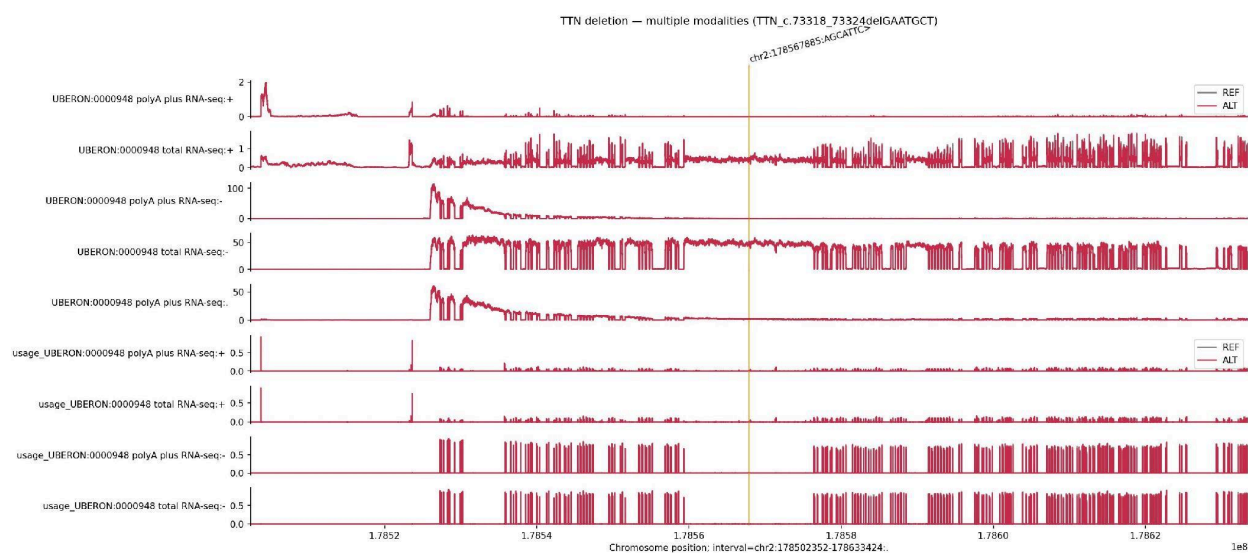


Figure 1. Multi-panel AlphaGenome output — TTN_c.73318_73324delGAATGCT — Heart tissue (UBERON:0000948). RNA-seq expression and splice site usage shown across 1 Mb window.

Panel 1: RNA-seq Expression (REF vs ALT)

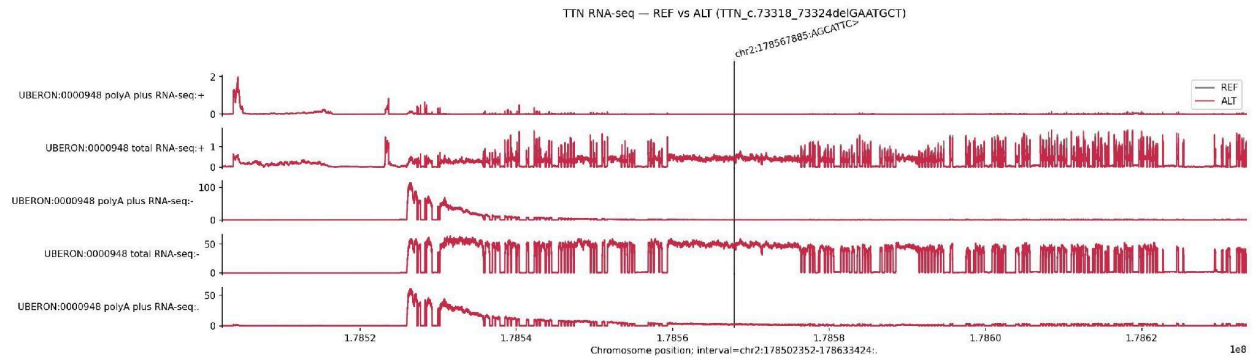


Figure 2. RNA-seq expression tracks — polyA-plus and total RNA-seq, both strands. REF (gray) and ALT (red) are nearly identical across the locus.

What This Shows

The REF and ALT expression tracks are nearly identical across the full TTN locus. At first glance this may appear to suggest the deletion has no effect, but this finding is expected and interpretively important:

- This is a heterozygous variant. The patient retains one functional TTN allele, which continues to produce normal transcripts.
- AlphaGenome models transcription and RNA output. The pathogenic mechanism of a frameshift truncation is at the protein level — the truncated titin protein is non-functional, but the transcriptional signal from the remaining normal allele is preserved.
- Absence of allele-specific expression change in RNA-seq output is consistent with a loss-of-function truncating variant where haploinsufficiency (insufficient titin from one allele) is the disease mechanism, rather than a dominant-negative effect.

Panel 2: Splice Site Usage (REF vs ALT)

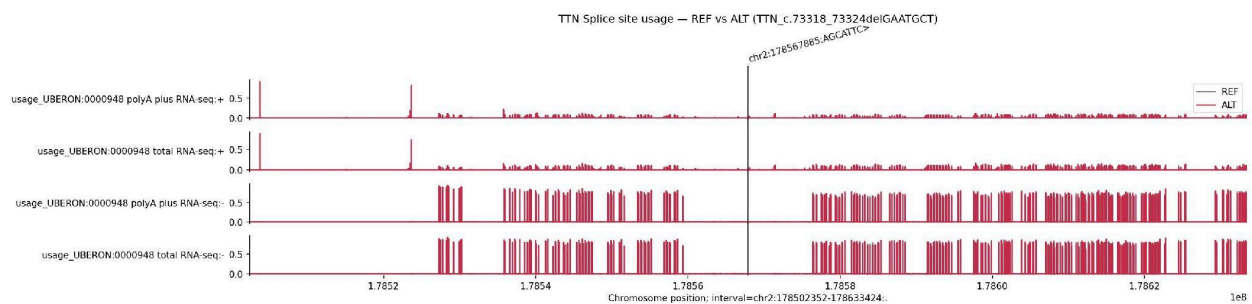


Figure 3. Splice site usage tracks across the TTN locus. Dense alternating pattern reflects TTN's extreme alternative splicing complexity in cardiac tissue.

What This Shows

The splice site usage pattern across the TTN locus is extraordinarily dense, reflecting the biology of TTN itself: it is one of the most alternatively spliced genes in the human genome.

Cardiac and skeletal muscle isoforms differ substantially in their splicing patterns, and the cardiac N2B and N2BA isoforms have distinct exon inclusion profiles that determine the passive stiffness properties of the sarcomere.

REF and ALT splice usage tracks remain closely matched throughout. This indicates the deletion is not creating a cryptic splice site or disrupting nearby splice regulatory elements. The frameshift mechanism is consistent: the deletion simply shifts the reading frame and introduces a premature stop, rather than altering splicing architecture. This is mechanistically reassuring — the pathogenic effect is contained and predictable.

Panel 3: Chromatin Accessibility (ATAC-seq)

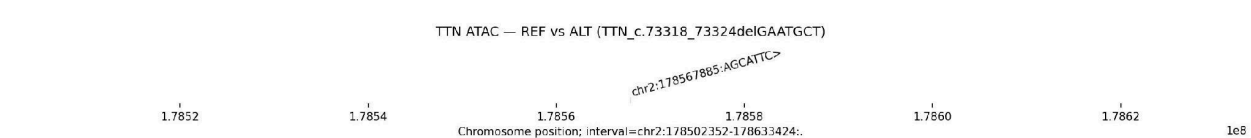


Figure 4. ATAC-seq chromatin accessibility — REF vs ALT. Note: output appears minimal in this run.

What This Shows

Data Gap Note

The ATAC-seq panel rendered with minimal visible signal in this AlphaGenome run. This is likely a model output or rendering limitation rather than a true biological finding — the TTN locus in cardiac tissue is expected to show chromatin accessibility signal. This panel should be considered incomplete and is not interpretable as a meaningful result. It is included for completeness and transparency.

Interpretation Summary

Taken together, the AlphaGenome outputs are consistent with the expected molecular behavior of a heterozygous TTN frameshift truncation in cardiac tissue:

Modality	Finding	Interpretation
RNA-seq Expression	REF and ALT tracks nearly identical	Expected — functional allele preserved; pathogenicity is protein-level
Splice Site Usage	No disruption of splice architecture	Consistent with clean frameshift; no cryptic splicing complication

ATAC-seq	Minimal signal rendered	Data gap — not interpretable; likely model output limitation
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Clinical Relevance for Mayo Evaluation

These computational outputs do not change the confirmed pathogenic classification of this variant — that determination was made through standard clinical genetic testing. Rather, they provide a tissue-specific view of how this deletion behaves at the TTN locus in a cardiac expression context, and confirm that the mechanism is consistent with haploinsufficiency via protein truncation. This is supplementary evidence supporting the dual-hit hypothesis: a known structural cardiac substrate (TTN truncating variant) interacting with an active infiltrative eosinophilic process to produce the observed multi-system phenotype.

Methods Note

Analysis performed using the Google DeepMind AlphaGenome API (Python SDK). Variant modeled as a deletion at hg38 chr2:178567885 (AGCATTC > deletion). Predictions generated for a 1 Mb genomic window centered on the variant position. Tissue ontology: UBERON:0000948 (heart). All outputs represent model predictions and have not been validated against experimental functional data for this specific variant.

Source files: ttn_rna_seq_ref_alt.png, ttn_splice_site_usage_ref_alt.png, ttn_atac_ref_alt.png, ttn_multi_panel_ref_alt.png